Eighth Grade Tile Task

Standard 8.NS.1

Name: _____ Date: _____

8

1. **a)** Tenths digit of 0.444...

b) Tenths digit of 0.999...

c) Square root of 0

2. **a)** Tenths digit of 0.222...

b) Square root of 1

c) Square root of 36

3. **a)** Square root of 9

b) Square root of 49

c) Square root of 25

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.NS.2

Name: _____ Date: _____

8

1. **a)** Whole part of $\sqrt{0}$ ☐

b) Whole part of $\sqrt{81}$ ☐

c) Whole part of $\sqrt{47}$ ☐

2. **a)** Whole part of $\sqrt{15}$ ☐

b) Whole part of $\sqrt{33}$ ☐

c) Whole part of $\sqrt{7}$ ☐

3. **a)** Whole part of $\sqrt{2}$ ☐

b) Whole part of $\sqrt{57}$ ☐

c) Whole part of $\sqrt{79}$ ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.EE.1

Name: _____ Date: _____

8

1. **a)** x^2 × x^4 = x^?: build the exponent ☐

   b) x^5 × x^4 = x^?: build the exponent ☐

   c) x^0 × x^7 = x^?: build the exponent ☐

2. **a)** x^1 × x^0 = x^?: build the exponent ☐

   b) x^2 × x^0 = x^?: build the exponent ☐

   c) x^0 × x^0 = x^?: build the exponent ☐

3. **a)** x^1 × x^3 = x^?: build the exponent ☐

   b) x^5 × x^0 = x^?: build the exponent ☐

   c) x^7 × x^1 = x^?: build the exponent ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.2

Name: _____ Date: _____

8

1. **a)** Square root of 25 ☐
   b) Square root of 16 ☐
   c) Square root of 1 ☐

2. **a)** Square root of 9 ☐
   b) Square root of 49 ☐
   c) Cube root of 8 ☐

3. **a)** Square root of 81 ☐
   b) Square root of 36 ☐
   c) Cube root of 0 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.3

Name: _____ Date: _____

8

1. **a)** Exponent in 1 × 10^k

b) Exponent in 2 × 10^k

c) Exponent in 9 × 10^k

2. **a)** Exponent in 5 × 10^k

b) Exponent in 5 × 10^k

c) Exponent in 4 × 10^k

3. **a)** Exponent in 5 × 10^k

b) Exponent in 8 × 10^k

c) Exponent in 5 × 10^k

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.EE.4

Name: _____ Date: _____

8

1. a) Exponent in 6 × 10^k for the number 60000000 ☐

b) Exponent in 3 × 10^k for the number 300000 ☐

c) Exponent in 9 × 10^k for the number 9000 ☐

2. a) Exponent in 7 × 10^k for the number 700000000 ☐

b) Exponent in 6 × 10^k for the number 600 ☐

c) Exponent in 6 × 10^k for the number 60000 ☐

3. a) Exponent in 8 × 10^k for the number 8000000 ☐

b) Exponent in 7 × 10^k for the number 7000000000 ☐

c) Exponent in 7 × 10^k for the number 70 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.EE.5

Name: _____ Date: _____

8

1. **a)** Slope of y = 2x + 2 ☐

b) Slope of y = 8x + 3 ☐

c) Slope of y = 5x + 1 ☐

2. **a)** Slope of y = 9x + 1 ☐

b) Slope of y = 6x + 3 ☐

c) Slope of y = 7x + 3 ☐

3. **a)** Slope of y = 1x + 3 ☐

b) Slope of y = 3x + 2 ☐

c) Slope of y = 0x + 1 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.EE.6

Name: _____ Date: _____

8

1. a) y-intercept of y = 3x + 2 ☐

b) y-intercept of y = 0x + 9 ☐

c) y-intercept of y = 3x + 0 ☐

2. a) y-intercept of y = 1x + 8 ☐

b) y-intercept of y = 1x + 5 ☐

c) y-intercept of y = 2x + 7 ☐

3. a) y-intercept of y = 1x + 1 ☐

b) y-intercept of y = 4x + 3 ☐

c) y-intercept of y = 3x + 4 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.EE.7

Name: _____ Date: _____

8

1. a) Solve 5x + 8 = 28: x = ? ☐

b) Solve 2x + 5 = 11: x = ? ☐

c) Solve 4x + 6 = 14: x = ? ☐

2. a) Solve 9x + 9 = 90: x = ? ☐

b) Solve 4x + 0 = 24: x = ? ☐

c) Solve 4x + 9 = 29: x = ? ☐

3. a) Solve 4x + 8 = 8: x = ? ☐

b) Solve 6x + 3 = 51: x = ? ☐

c) Solve 4x + 7 = 35: x = ? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.EE.8

Name: _____ Date: _____

8

1. a) Solve x + y = 14, x - y = 0: x = ? ☐

b) Solve x + y = 1, x - y = 1: x = ? ☐

c) Solve x + y = 5, x - y = 1: x = ? ☐

2. a) Solve x + y = 6, x - y = 4: x = ? ☐

b) Solve x + y = 0, x - y = 0: x = ? ☐

c) Solve x + y = 15, x - y = 3: x = ? ☐

3. a) Solve x + y = 7, x - y = 1: x = ? ☐

b) Solve x + y = 11, x - y = 1: x = ? ☐

c) Solve x + y = 3, x - y = 1: x = ? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.1

Name: _____ Date: _____

8

1. a) f(x) = 3x + 2. Find f(0). ☐

b) f(x) = 1x + 2. Find f(1). ☐

c) f(x) = 1x + 1. Find f(0). ☐

2. a) f(x) = 2x + 0. Find f(0). ☐

b) f(x) = 1x + 5. Find f(2). ☐

c) f(x) = 3x + 0. Find f(2). ☐

3. a) f(x) = 2x + 4. Find f(0). ☐

b) f(x) = 2x + 3. Find f(3). ☐

c) f(x) = 1x + 5. Find f(0). ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.F.2

Name: _____ Date: _____

8

1. a) Slope of function A: y = 6x

b) Slope of function A: y = 4x

c) Slope of function A: y = 3x

2. a) Slope of function A: y = 8x

b) Slope of function A: y = 2x

c) Slope of function A: y = 0x

3. a) Slope of function A: y = 7x

b) Slope of function A: y = 1x

c) Slope of function A: y = 9x

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.3

Name: _____ Date: _____

8

1. **a)** $f(x) = x^2 + 5$. Find $f(1)$. ☐

b) $f(x) = x^2 + 2$. Find $f(1)$. ☐

c) $f(x) = x^2 + 7$. Find $f(0)$. ☐

2. **a)** $f(x) = x^2 + 8$. Find $f(0)$. ☐

b) $f(x) = x^2 + 4$. Find $f(0)$. ☐

c) $f(x) = x^2 + 0$. Find $f(0)$. ☐

3. **a)** $f(x) = x^2 + 1$. Find $f(1)$. ☐

b) $f(x) = x^2 + 8$. Find $f(1)$. ☐

c) $f(x) = x^2 + 1$. Find $f(2)$. ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.4

Name: _____ Date: _____

8

1. **a)** Rate of change in y = 3x + 2 ☐

   b) Rate of change in y = 9x + 1 ☐

   c) Rate of change in y = 4x + 2 ☐

2. **a)** Rate of change in y = 1x + 1 ☐

   b) Rate of change in y = 7x + 1 ☐

   c) Rate of change in y = 6x + 1 ☐

3. **a)** Rate of change in y = 0x + 0 ☐

   b) Rate of change in y = 5x + 2 ☐

   c) Rate of change in y = 2x + 2 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.F.5

Name: _____ Date: _____

8

1. **a)** y-intercept of a line through (0, 0) ☐
   b) y-intercept of a line through (0, 5) ☐
   c) y-intercept of a line through (0, 2) ☐

2. **a)** y-intercept of a line through (0, 9) ☐
   b) y-intercept of a line through (0, 1) ☐
   c) y-intercept of a line through (0, 3) ☐

3. **a)** y-intercept of a line through (0, 4) ☐
   b) y-intercept of a line through (0, 8) ☐
   c) y-intercept of a line through (0, 7) ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.1

Name: _____ Date: _____

8

1. a) Reflect (3, 1) across the x-axis: new y is? ☐

b) Reflect (3, 3) across the x-axis: new y is? ☐

c) Reflect (3, 7) across the x-axis: new y is? ☐

2. a) Reflect (3, 6) across the x-axis: new y is? ☐

b) Reflect (3, 0) across the x-axis: new y is? ☐

c) Reflect (3, 5) across the x-axis: new y is? ☐

3. a) Reflect (3, 8) across the x-axis: new y is? ☐

b) Reflect (3, 2) across the x-axis: new y is? ☐

c) Reflect (3, 9) across the x-axis: new y is? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.2

Name: _____ Date: _____

8

1. a) Sides on a pentagon? ☐

b) Sides on a square? ☐

c) Sides on a heptagon? ☐

2. a) Sides on an octagon? ☐

b) Sides on a hexagon? ☐

c) Intersection points of two parallel lines? ☐

3. a) Sides on a nonagon? ☐

b) Intersection points of two distinct lines? ☐

c) Lines needed to form a triangle? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.3

Name: _____ Date: _____

8

1. **a)** Dilate a 3 cm segment by 3: new length? ☐

b) Dilate a 1 cm segment by 1: new length? ☐

c) Dilate a 2 cm segment by 1: new length? ☐

2. **a)** Dilate a 7 cm segment by 1: new length? ☐

b) Dilate a 2 cm segment by 2: new length? ☐

c) Dilate a 1 cm segment by 8: new length? ☐

3. **a)** Dilate a 2 cm segment by 3: new length? ☐

b) Dilate a 5 cm segment by 1: new length? ☐

c) Dilate a 3 cm segment by 1: new length? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.G.4

Name: _____ Date: _____

8

1. **a)** Scale factor from 2 to 16: build the factor ☐
   b) Scale factor from 2 to 6: build the factor ☐
   c) Scale factor from 2 to 10: build the factor ☐

2. **a)** Scale factor from 2 to 18: build the factor ☐
   b) Scale factor from 2 to 4: build the factor ☐
   c) Scale factor from 2 to 12: build the factor ☐

3. **a)** Scale factor from 2 to 14: build the factor ☐
   b) Scale factor from 2 to 2: build the factor ☐
   c) Scale factor from 2 to 8: build the factor ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.5

Name: _____ Date: _____

8

1. **a)** Supplementary angle of 177°

   b) Supplementary angle of 178°

   c) Supplementary angle of 179°

2. **a)** Supplementary angle of 171°

   b) Supplementary angle of 180°

   c) Supplementary angle of 176°

3. **a)** Supplementary angle of 172°

   b) Supplementary angle of 175°

   c) Complementary angle of 83°

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.G.6

Name: _____ Date: _____

8

1. **a)** Value of $1^2 + 1^2$? ☐

b) Value of $3^2 - 2^2$? ☐

c) Value of $4^2 - 3^2$? ☐

2. **a)** Value of $0^2 + 0^2$? ☐

b) Value of $2^2 - 1^2$? ☐

c) Value of $2^2 + 2^2$? ☐

3. **a)** Value of $2^2 + 0^2$? ☐

b) Value of $3^2 + 0^2$? ☐

c) Value of $1^2 + 1^2 + 2^2$? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.G.7

Name: _____ Date: _____

8

1. **a)** Area of a right triangle with legs 7 and 2? ☐

b) Area of a 3-4-5 right triangle? ☐

c) Area of a right triangle with legs 3 and 6? ☐

2. **a)** Area of a right triangle with legs 2 and 2? ☐

b) Area of a square with side 1? ☐

c) Hypotenuse 10, leg 6: other leg? ☐

3. **a)** Hypotenuse 5, leg 4: other leg? ☐

b) Area of a right triangle with legs 5 and 2? ☐

c) Area of a square with side 2? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.G.8

Name: _____ Date: _____

8

1. **a)** Dilate point at x = 1 by 5: new x? ☐

b) Dilate point at x = 1 by 2: new x? ☐

c) Dilate point at x = 4 by 1: new x? ☐

2. **a)** Dilate point at x = 3 by 1: new x? ☐

b) Dilate point at x = 1 by 1: new x? ☐

c) Dilate point at x = 7 by 1: new x? ☐

3. **a)** Dilate point at x = 1 by 8: new x? ☐

b) Dilate point at x = 9 by 1: new x? ☐

c) Dilate point at x = 3 by 2: new x? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.G.9

Name: _____ Date: _____

8

1. a) Volume of a 1 by 7 by 1 prism

b) Volume of a 2 by 4 by 1 prism

c) Volume of a 1 by 5 by 1 prism

2. a) Volume of a 1 by 2 by 1 prism

b) Volume of a 3 by 1 by 1 prism

c) Volume of a 4 by 1 by 1 prism

3. a) Volume of a cube with side 1

b) Volume of a cylinder, r=1, height 2 ($\pi \approx 3$)

c) Volume of a 9 by 1 by 1 prism

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.SP.1

Name: _____ Date: _____

8

1. **a)** Scatter plot with 7 data points: how many dots? ☐
   b) Scatter plot with 2 data points: how many dots? ☐
   c) Scatter plot with 4 data points: how many dots? ☐

2. **a)** Scatter plot with 3 data points: how many dots? ☐
   b) Scatter plot with 8 data points: how many dots? ☐
   c) Scatter plot with 1 data points: how many dots? ☐

3. **a)** Scatter plot with 9 data points: how many dots? ☐
   b) Scatter plot with 5 data points: how many dots? ☐
   c) Scatter plot with 6 data points: how many dots? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Eighth Grade Tile Task

Standard 8.SP.2

Name: _____ Date: _____

8

1. a) Best-fit line y = 8x: predicted y at x = 1 ☐

b) Best-fit line y = 2x: predicted y at x = 1 ☐

c) Best-fit line y = 6x: predicted y at x = 1 ☐

2. a) Best-fit line y = 0x: predicted y at x = 1 ☐

b) Best-fit line y = 5x: predicted y at x = 1 ☐

c) Best-fit line y = 1x: predicted y at x = 1 ☐

3. a) Best-fit line y = 9x: predicted y at x = 1 ☐

b) Best-fit line y = 3x: predicted y at x = 1 ☐

c) Best-fit line y = 7x: predicted y at x = 1 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: EIGHTH GRADE

Eighth Grade Tile Task

Standard 8.SP.3

Name: _____ Date: _____

8

1. **a)** Slope in y = 2x + 0 ☐

b) Slope in y = 4x + 1 ☐

c) Slope in y = 1x + 1 ☐

2. **a)** Slope in y = 3x + 1 ☐

b) Slope in y = 0x + 1 ☐

c) Slope in y = 8x + 0 ☐

3. **a)** Slope in y = 5x + 0 ☐

b) Slope in y = 6x + 0 ☐

c) Slope in y = 9x + 0 ☐

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Eighth Grade Tile Task

Standard 8.SP.4

Name: _____ Date: _____

8

1. a) Two-way table: 1 boys and 0 girls. Total? ☐

b) Two-way table: 3 boys and 3 girls. Total? ☐

c) Two-way table: 0 boys and 8 girls. Total? ☐

2. a) Two-way table: 2 boys and 1 girls. Total? ☐

b) Two-way table: 0 boys and 2 girls. Total? ☐

c) Two-way table: 0 boys and 5 girls. Total? ☐

3. a) Two-way table: 4 boys and 0 girls. Total? ☐

b) Two-way table: 1 boys and 8 girls. Total? ☐

c) Two-way table: 1 boys and 6 girls. Total? ☐

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9